

Data Dictionary

1. People

ACCOUNTS

One row is one login identity for a person who can sign in to Aria.

Attribute	Description	Data type / format	Key	Validation rules	Sample / default
account_id	Unique ID for the login account; shared key for the customer and staff subtypes	bigserial (generated)	PK	NOT NULL, UNIQUE	1001 (generated)
email	Login and contact email	varchar(320)	—	NOT NULL, UNIQUE; "@" must appear after the first character	diver@example.com
username	Unique login name	varchar(25)	—	NOT NULL, UNIQUE	reef_rider
password_hash	Hashed password; the plain password is never stored	varchar(255)	—	NOT NULL	\$2b\$12\$Kx...
created_at	When the account was created	timestampz (UTC)	—	NOT NULL	Default: now()

CUSTOMERS

One row is one account that buys from Aria (subtype of ACCOUNTS).

Attribute	Description	Data type / format	Key	Validation rules	Sample / default
customer_id	Same value as the customer's account ID	bigint	PK, FK → ACCOUNTS.account_id	NOT NULL; must exist in ACCOUNTS	1001
first_name	Customer's first name	varchar(100)	—	NOT NULL	Ana
last_name	Customer's last name	varchar(100)	—	NOT NULL	Lopez
phone	Contact phone number	varchar(20)	—	Optional	403-555-0142
is_active	Whether the record is in use; false hides it without deleting history	boolean	—	NOT NULL	Default: TRUE
created_at	When the row was created	timestampz	—	NOT NULL	Default: now()
updated_at	When the row was last changed (refreshed by trigger)	timestampz	—	NOT NULL	Default: now()

STAFF

One row is one Aria employee (subtype of ACCOUNTS).

Attribute	Description	Data type / format	Key	Validation rules	Sample / default
account_id	Same value as the employee's account ID	bigint	PK, FK → ACCOUNTS.account_id	NOT NULL; must exist in ACCOUNTS	2007
employee_no	Unique employee number	varchar(40)	—	NOT NULL, UNIQUE	E-0042
department	Department the employee works in (for example fulfilment, customer service, finance, delivery)	varchar(80)	—	NOT NULL	Fulfilment
hire_date	Date the employee was hired	date, YYYY-MM-DD	—	NOT NULL	2024-03-15
is_active	Whether the record is in use	boolean	—	NOT NULL	Default: TRUE

ADDRESSES

One row is one shipping and/or billing address saved by a customer.

Attribute	Description	Data type / format	Key	Validation rules	Sample / default
address_id	Unique ID for the address	bigserial (generated)	PK	NOT NULL, UNIQUE; UNIQUE together with account_id	301 (generated)
account_id	Customer who owns the address	bigint	FK → CUSTOMERS.customer_id	NOT NULL	1001
line1	First line of the street address	varchar(255)	—	NOT NULL	123 Harbour St
line2	Suite or unit number	varchar(255)	—	Optional	Unit 4
city	City	varchar(100)	—	NOT NULL	Calgary
region	State or province	varchar(100)	—	NOT NULL	Alberta
postal_code	Postal or ZIP code	varchar(20)	—	NOT NULL	T2P 1J9
country_code	Two-letter country code	char(2)	—	NOT NULL; one of CA, US	CA
is_shipping	Address can be	boolean	—	NOT NULL; at least one	Default: FALSE

	used for shipping			of is_shipping and is_billing must be TRUE	
is_billing	Address can be used for billing	boolean	—	NOT NULL; see is_shipping	Default: FALSE
is_default	This is the customer's default address	boolean	—	NOT NULL; at most one TRUE per customer	Default: FALSE
created_at	When the row was created	timestampz	—	NOT NULL	Default: now()

2. Catalog

CATEGORIES

One row is one product category, which may sit under a parent category.

Attribute	Description	Data type / format	Key	Validation rules	Sample / default
category_id	Unique ID for the category	bigserial (generated)	PK	NOT NULL, UNIQUE	12 (generated)
name	Category name	varchar(120)	—	NOT NULL, UNIQUE	Regulators
parent_category_id	Parent category; null for a	bigint	FK → CATEGORIES.category_id	Optional; cannot equal	3

	top-level category			category_id	
description	Short description) of the category	varchar(200	—	Optional	Breathing regulators and octopus sets

PRODUCTS

One row is one diving or watersports product for sale.

Attribute	Description	Data type / format	Key	Validation rules	Sample / default
product_id	Unique ID for the product	bigserial (generated)	PK	NOT NULL, UNIQUE	5001 (generated)
category_id	Category the product belongs to	bigint	FK → CATEGORIES.category_id	Optional (null if uncategorized)	12
sku	Stock-keeping unit code	varchar(50)	—	NOT NULL, UNIQUE	REG-APX-01
name	Product name	varchar(255)	—	NOT NULL	Apex Regulator Set
description	Full product description	text	—	Optional	Balanced diaphragm regulator...
unit_price	Current list price per unit	numeric(10,2)	—	NOT NULL; >= 0	349.99

currency	Currency of the price	currency_code (char(3))	—	NOT NULL; one of CAD, USD	Default: CAD
stock_quantity	Units on hand	integer	—	NOT NULL; >= 0	Default: 0
reorder_level	Stock level at which to reorder	integer	—	NOT NULL; >= 0	Default: 10
is_active	Whether the product is listed	boolean	—	NOT NULL	Default: TRUE
created_at	When the row was created	timestampz	—	NOT NULL	Default: now()
updated_at	When the row was last changed (refreshed by trigger)	timestampz	—	NOT NULL	Default: now()

3. Vendors

PAYEE

One row is one party Aria pays, either a carrier or a supplier.

Attribute	Description	Data type / format	Key	Validation rules	Sample / default
payee_id	Unique ID for the payee	bigserial (generated)	PK	NOT NULL, UNIQUE	40 (generated)

payee_type	Which subtype the payee belongs to	varchar(16)	—	NOT NULL; one of CARRIER, SUPPLIER	CARRIER
name	Payee name	varchar(120)	—	NOT NULL, UNIQUE	Northern Freight
created_at	When the row was created	timestampz	—	NOT NULL	Default: now()

CARRIERS

One row is one shipping company that delivers orders (subtype of PAYEE).

Attribute	Description	Data type / format	Key	Validation rules	Sample / default
carrier_id	Same value as the payee ID	bigint	PK, FK → PAYEE.payee_id	NOT NULL; payee_type must be CARRIER	40
tracking_url_template	Tracking link pattern with a {tracking_number} placeholder	varchar(2048)	—	Optional	https://track.example.com/{tracking_number}

SUPPLIERS

One row is one company Aria buys stock from (subtype of PAYEE).

Attribute	Description	Data type / format	Key	Validation rules	Sample / default
supplier_id	Same value as the payee ID	bigint	PK, FK → PAYEE.payee_id	NOT NULL; payee_type must be SUPPLIER	41
contact_email	Supplier contact email	varchar(254)	—	Optional	sales@oceangear.example.com

4. Sales

ORDERS

One row is one customer order, shipped as one shipment (partitioned monthly by order_date).

Attribute	Description	Data type / format	Key	Validation rules	Sample / default
order_id	Unique ID for the order	bigserial (generated)	PK	NOT NULL	8001 (generated)
order_date	When the order was placed; also the partition key	timestampz	PK	NOT NULL	Default: now()
customer_id	Customer who placed the order	bigint	FK → CUSTOMERS.customer_id	NOT NULL	1001

shipping_address_id	Address the order ships to	bigint	FK → ADDRESSES (with customer_id)	NOT NULL; must belong to the same customer	301
billing_address_id	Address the order is billed to	bigint	FK → ADDRESSES (with customer_id)	NOT NULL; must belong to the same customer	301
carrier_id	Carrier delivering the order	bigint	FK → CARRIERS.carrier_id	Optional (null until assigned)	40
status	Stage in the fulfilment process	varchar(16)	—	NOT NULL; one of PENDING, PAID, PROCESSING, SHIPPED, DELIVERED, CANCELLED, REFUNDED	Default: PENDING
currency	Currency of all amounts on the order	currency_code (char(3))	—	NOT NULL; one of CAD, USD	CAD
tracking_number	Carrier tracking number	varchar(100)	—	Optional (null until shipped)	1Z999AA10123456784

subtotal	Sum of line totals before shipping and tax	numeric(10,2)	—	NOT NULL; >= 0	349.99
shipping_cost	Shipping charge	numeric(10,2)	—	NOT NULL; >= 0	Default: 0
tax_amount	Tax charged	numeric(10,2)	—	NOT NULL; >= 0	Default: 0
total_amount	Subtotal + shipping_cost + tax_amount (generated, cannot be edited)	numeric(10,2), stored generated	—	Derived	384.99
updated_at	When the row was last changed (refreshed by trigger)	timestampz	—	NOT NULL	Default: now()

ORDER_ITEMS

One row is one product line on an order (partitioned monthly by order_date).

Attribute	Description	Data type / format	Key	Validation rules	Sample / default
order_item_id	Unique ID for the order line	bigserial (generated)	PK	NOT NULL	90001 (generated)

order_date	Copy of the order's order_date; partition key	timestamptz	PK, FK → ORDERS (with order_id)	NOT NULL	2026-09-14 10:32:00+00
order_id	Order the line belongs to	bigint	FK → ORDERS.order_id	NOT NULL; UNIQUE with order_date and product_id	8001
product_id	Product sold on the line	bigint	FK → PRODUCTS.product_id	NOT NULL	5001
quantity	Units sold	integer	—	NOT NULL; > 0	2
unit_price	Price per unit at the time of sale	numeric(10,2)	—	NOT NULL; >= 0	349.99
line_total	quantity × unit_price (generated)	numeric(10,2), stored generated	—	Derived	699.98

5. Payments and refunds

PAYMENTS

One row is one payment made toward an order (partitioned monthly by order_date).

Attribute	Description	Data type / format	Key	Validation rules	Sample / default
payment_id	Unique ID for the payment	bigserial (generated)	PK	NOT NULL	7001 (generated)

order_date	Copy of the order's order_date; partition key	timestampz	PK, FK → ORDERS (with order_id)	NOT NULL	2026-09-14 10:32:00+00
order_id	Order the payment is for	bigint	FK → ORDERS.order_id	NOT NULL	8001
amount	Amount paid	numeric(10,2)	—	NOT NULL; >= 0	384.99
currency	Currency of the payment	currency_code (char(3))	—	NOT NULL; one of CAD, USD	CAD
payment_method	Method used; decides which subtype table holds the details	varchar(20)	—	NOT NULL; one of CREDIT_CARD, DEBIT_CARD, PAYPAL, GIFT_CARD	CREDIT_CARD
processed_at	When the payment processor handled the payment	timestampz	—	Optional	2026-09-14 10:33:12+00
status	Payment outcome	varchar(20)	—	NOT NULL; one of PENDING, APPROVED, DECLINED, REFUNDED	Default: PENDING

transaction_ref	Reference from the external payment processor	varchar(30)	—	NOT NULL	TXN-20260914-0042
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CARD_PAYMENTS

One row is the card details for one credit or debit card payment (subtype of PAYMENTS).

Attribute	Description	Data type / format	Key	Validation rules	Sample / default
payment_id	Payment these details belong to	bigint	PK, FK → PAYMENTS (with order_date)	NOT NULL; parent method must be CREDIT_CARD or DEBIT_CARD	7001
order_date	Copy of the payment's order_date	timestamp tz	PK, FK → PAYMENTS	NOT NULL	2026-09-14 10:32:00+00
last4	Last four digits of the card; full number is never stored	char(4), digits only	—	NOT NULL; must match 4 digits	4242
brand	Card brand	varchar(30)	—	NOT NULL	Visa
auth_code	Authorization code from the card processor	varchar(80)	—	NOT NULL	A1B2C3

PAYPAL_PAYMENTS

One row is the PayPal details for one PayPal payment (subtype of PAYMENTS).

Attribute	Description	Data type / format	Key	Validation rules	Sample / default
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payment_id	Payment these details belong to	bigint	PK, FK → PAYMENTS (with order_date)	NOT NULL; parent method must be PAYPAL	7002
order_date	Copy of the payment's order_date	timestampz	PK, FK → PAYMENTS	NOT NULL	2026-09-15 08:10:00+00
payer_id	PayPal's ID for the payer	varchar(120)	—	NOT NULL	5K7QX9ABCD123
payer_email	Email on the PayPal account	varchar(320)	—	NOT NULL	ana@example.com

GIFT_CARD_PAYMENTS

One row is the gift card details for one gift card payment (subtype of PAYMENTS).

Attribute	Description	Data type / format	Key	Validation rules	Sample / default
payment_id	Payment these details belong to	bigint	PK, FK → PAYMENTS (with order_date)	NOT NULL; parent method must be GIFT_CARD	7003
order_date	Copy of the payment's order_date	timestampz	PK, FK → PAYMENTS	NOT NULL	2026-09-16 14:45:00+00

gift_card_id	ID of the gift card redeemed	varchar(80)	—	NOT NULL	GC-8841-2207
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REFUNDS

One row is one refund request against a payment.

Attribute	Description	Data type / format	Key	Validation rules	Sample / default
refund_id	Unique ID for the refund	bigserial (generated)	PK	NOT NULL, UNIQUE	600 (generated)
payment_id	Payment being refunded	bigint	FK → PAYMENTS (with payment_order_date)	NOT NULL	7001
payment_order_date	order_date of the payment (part of the composite FK)	timestamp tz	FK → PAYMENTS	NOT NULL	2026-09-14 10:32:00+00
approved_by	Staff member who approved the refund	bigint	FK → STAFF.account_id	Required unless status is REQUESTED	2007
amount	Amount refunded	numeric(10,2)	—	NOT NULL; >= 0	120.00
currency	Currency of the refund	currency_code (char(3))	—	NOT NULL; one of CAD, USD	CAD
reason	Reason for the refund	varchar(255)	—	Optional	Item arrived damaged

status	Refund stage	varchar(20)	—	NOT NULL; one of REQUESTED, APPROVED, DENIED, COMPLETED	Default: REQUESTED
requested_at	When the refund was requested	timestampz	—	NOT NULL	Default: now()
resolved_at	When the refund was approved, denied or completed	timestampz	—	Optional; must be >= requested_at	NULL while open

REFUND_ITEMS

One row is the part of one order line that is refunded under one refund.

Attribute	Description	Data type / format	Key	Validation rules	Sample / default
refund_id	Refund the row belongs to	bigint	PK, FK → REFUNDS.refund_id	NOT NULL	600
order_item_id	Order line being refunded	bigint	PK, FK → ORDER_ITEMS (with order_date)	NOT NULL	90001

order_date	Copy of the order line's order_date	timestampz	PK, FK → ORDER_ITEMS	NOT NULL	2026-09-14 10:32:00+00
quantity	Units refunded from the line	integer	—	NOT NULL; > 0	1
amount	Amount refunded for the line	numeric(10,2)	—	NOT NULL; >= 0	120.00

6. Finance

ACCOUNTS_RECEIVABLE

One row is one amount owed to Aria for an order.

Attribute	Description	Data type / format	Key	Validation rules	Sample / default
ar_id	Unique ID for the receivable	bigserial (generated)	PK	NOT NULL, UNIQUE	21 (generated)
order_id	Order the money is owed for	bigint	FK → ORDERS (with order_date)	NOT NULL	8001
order_date	Copy of the order's order_date	timestampz	FK → ORDERS	NOT NULL	2026-09-14 10:32:00+00
amount_due	Amount owed	numeric(10,2)	—	NOT NULL; >= 0	384.99
currency	Currency of the amount	currency_code (char(3))	—	NOT NULL; one of CAD, USD	CAD
due_date	Date payment is due	timestampz	—	NOT NULL	2026-10-14 00:00:00+00

status	Collection stage	varchar(20)	—	NOT NULL; one of OPEN, PAID, OVERDUE, WRITTEN_OFF	Default: OPEN
created_at	When the row was created	timestampz	—	NOT NULL	Default: now()

ACCOUNTS_PAYABLE

One row is one amount Aria owes to a carrier or supplier.

Attribute	Description	Data type / format	Key	Validation rules	Sample / default
ap_id	Unique ID for the payable	bigserial (generated)	PK	NOT NULL, UNIQUE	15 (generated)
payee_id	Carrier or supplier Aria owes	bigint	FK → PAYEE.payee_id	NOT NULL	40
amount_due	Amount owed	numeric(10,2)	—	NOT NULL; >= 0	1250.00
currency	Currency of the amount	currency_code (char(3))	—	NOT NULL; one of CAD, USD	CAD
due_date	Date payment is due	timestampz	—	NOT NULL	2026-10-31 00:00:00+00
status	Payment stage	varchar(20)	—	NOT NULL; one of OPEN, PAID, OVERDUE, WRITTEN_OFF	Default: OPEN

7. Chatbot knowledge base

POLICY_DOCUMENTS

One row is one policy document used as a source for the chatbot.

Attribute	Description	Data type / format	Key	Validation rules	Sample / default
document_id	Unique ID for the policy document	bigserial (generated)	PK	NOT NULL, UNIQUE	3 (generated)
title	Document title	varchar(240)	—	NOT NULL	Returns and Refunds Policy
version	Version label	varchar(40)	—	NOT NULL	Default: 1.0
last_updated	When the document was last revised	timestampz	—	NOT NULL	Default: now()
category	Topic, such as Shipping, Refunds or HR	varchar(50)	—	Optional	Refunds
file_path	Location of the source file used for ingestion	varchar(512)	—	Optional	/policies/returns_v1.pdf
is_active	Whether the document is in use	boolean	—	NOT NULL	Default: TRUE

POLICY_DOCUMENT_CHUNKS

One row is one text chunk of a policy document, stored with its embedding for retrieval.

Attribute	Description	Data type / format	Key	Validation rules	Sample / default
chunk_id	Unique ID for the chunk	bigserial (generated)	PK	NOT NULL, UNIQUE	140 (generated)

document_id	Document the chunk came from	bigint	FK → POLICY_DOCUMENTS.document_id	NOT NULL	3
chunk_index	Position of the chunk within the document	integer	—	NOT NULL; >= 0; UNIQUE with document_id	0
chunk_text	Text of the chunk	text	—	NOT NULL	Items may be returned within 30 days...
embedding	1536-dimension vector used for similarity search	vector(1536) (pgvector)	—	NOT NULL	[0.012, -0.044, ...]